Systems Thinking for Better Science #2: **Introduction to Gen AI Workflows**

Setting Aside the Hype and the Fear, and Getting Started

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Link to presentation: https://github.com/dsih-artpark/workshops/blob/main/workshops/workshop-02-gen-ai-workflows-01/presentation.pdf

Raise your hand if AI has surprised you - in either direction.

Liz Fosslien [1]

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Trust me, I'm NOT an expert either. I just like yapping.

This Isn't Specifically an AI Problem

New technology has always felt this way.

The spiral isn't a sign that something is wrong with you or the tools. It's the emotional arc of every significant shift in how we work.

So you have to first navigate your relationship with your work before you jump to the next step.

Even Start Ups and Enterprises Are Still Figuring It Out

The scale of adoption:

- Ramp: 99.5% of their team on AI tools, usage up 6,300% year-on-year [2]
- Shopify's CEO made AI proficiency a performance expectation for every employee [3]

What that's also produced:

- Amazon's main shopping site went down for six hours following AI-assisted code changes [4]
- 40% of workers have received AI-generated content that looks polished but communicates nothing - each instance costs roughly 2 hours to resolve [5]

The confusion is rational. Especially at the frontier, this is hard.

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- Which in turn drives major fears - fear of missing out, fear of being left behind

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Disclaimer: This talk contains what I've derived from my readings and experiences. YMMV[1].

[1] Your mileage may vary.

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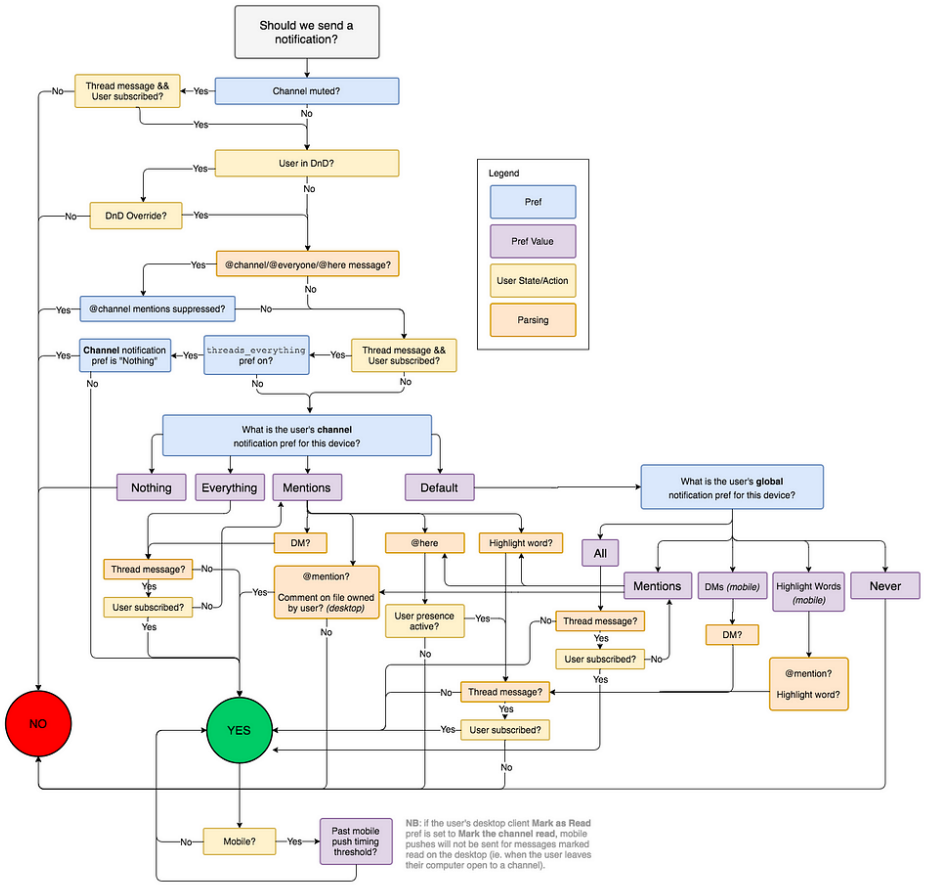
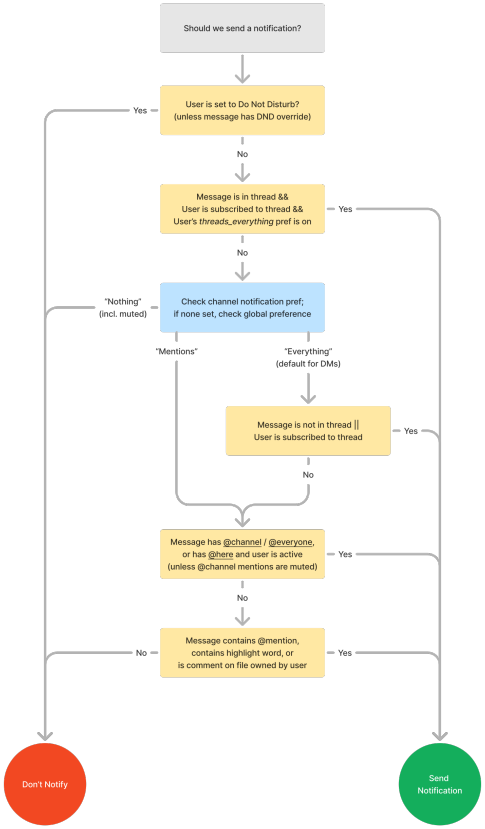
The thinking is the work. Delegation comes after. Think of AI as a talented junior you can delegate work to. She's talented, but she's a junior.

The Original Slack Flowchart



Slack, via Sophie Alpert [6]

After: Starting From What the User Actually Wants

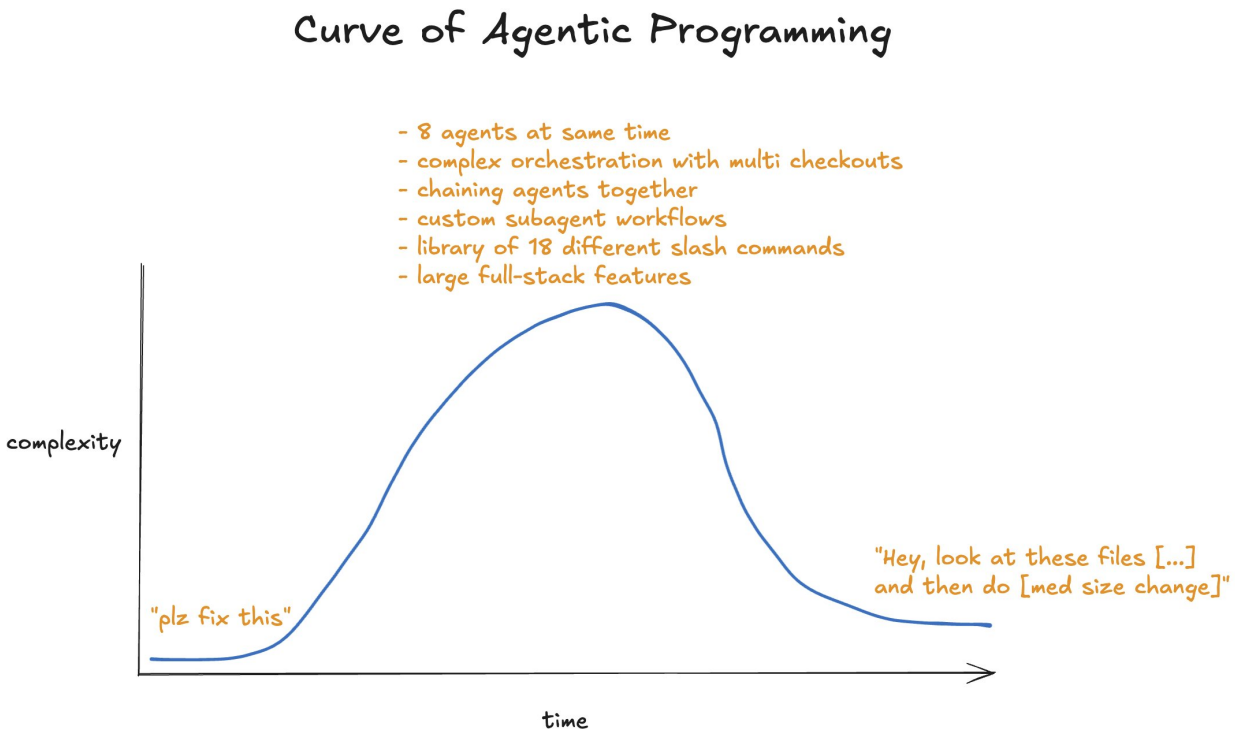


Sophie Alpert [6]

Next move: once the thinking is clear, the asking can be plain.

Simpler Inputs Work Better Over Time

KISS: Keep it Simple, Stupid



Peter Steinberger [7]

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If this could be automated and generalised, the frontier AI labs wouldn't be hiring more than ever.

← **Post**

Duca @big_duca · Feb 14

The thing I don't get is:

Claude Code is writing 100% of Claude code now.

But Anthropic has 100+ open dev positions on their jobs page.

?

458 134 5.1K 657K

Boris Cherny
@bcherny

Someone has to prompt the Claudes, talk to customers, coordinate with other teams, decide what to build next. Engineering is changing and great engineers are more important than ever.

1:28 AM · Feb 15, 2026 · **1.1M** Views

Boris Cherny on X [13]

#2: Ask Plainly After You've Thought

This is great news! It means your job is more secure than you thought. It also means all roles are evolving closer into product roles.
Describe what you want like you'd tell a colleague using direct and simple language, and iterate from there. (More on that in #3.)

Clear thinking first. Plain speech second.

This one's an original, no reference. Why're you still reading this?

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Writing is a product - and good writing means having empathy for your reader. It's all product design.

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Your core work isn't the writing itself, but what the writing accomplishes and who can understand it. Good writing takes time!

This one's an original as well. Your core work is often navigating that crash, with or without AI.

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Above all else, be curious. Read and explore. Ask lots of questions and share what you learn! *(Slack: #technical-support)*

Contradictions

Some of what makes you uneasy about AI isn't in your head.

- Code is cheaper to generate [11], but not cheaper to own.
- Fast output can still create real costs: compute, oversight, and technical debt.
- Labour-saving tools can also raise standards and expectations [9].

The question isn't whether AI makes work faster. It's what else comes with that speed.

Contradictions

A BCG study of 1,488 workers (March 2026) found [12]:

- Heavy AI oversight → 14% more mental effort, 33% more decision fatigue
- Workers described "buzzing," mental fog, slower thinking
- But: using AI to replace repetitive tasks reduced burnout by 15%

Technologies sold as time-saving often end up extending and intensifying work [10].

There is a cognitive load to using AI, and these are innate contradictions to this set of emerging technologies.

What I'd Hope You Take With You

- Think first, then delegate.
- Ask plainly after you've thought.
- Read everything you ship.

And accept that there are contradictions.

AI can make work faster and stranger at the same time. Useful and exhausting. Powerful and costly.

The goal is not to resolve all of that. It's to work with it more deliberately.

Stay In Touch

Link to presentation:

`https://github.com/dsih-artpark/workshops/blob/main/workshops/workshop-02-gen-ai-workflows-01/presentation.pdf`

Reach out if you have workflows you'd like to share.

Future workshops are dependent on your interest.

Slack Channel: #technical-support

Or DM.

Coming Up

Different team members can take up different workshop themes next:

1. Advanced Experiment Tracking & Documentation (with gen AI powered workflows)

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3. Writing and Implementing Engineering Specs, Project Tooling when using Code Gen

References & Reading

[1] Fosslien L (2024) The AI adoption curve often involves an emotional spiral [LinkedIn post]. LinkedIn. Available: https://www.linkedin.com/posts/liz-fosslien_the-ai-adoption-curve-often-involves-an-emotional-activity-7449860022924185600-Q2PZ/. Accessed 2026 Apr 15.

[2] Majic Predin J (2026) How To Get A Company AI Pilled And What VCs Want To See Next. Forbes. Available: https://www.forbes.com/sites/josipamajic/2026/04/09/how-to-get-a-company-ai-pilled-and-what-vcs-want-to-see-next/. Accessed 2026 Apr 15.

[3] Palmer A (2025) Shopify CEO says staffers need to prove jobs can't be done by AI before asking for more headcount. CNBC. Available: https://www.cnbc.com/2025/04/07/shopify-ceo-prove-ai-cant-do-jobs-before-asking-for-more-headcount.html. Accessed 2026 Apr 15.

[4] Morales J (2026) Amazon calls upon senior engineers to address issues created by "Gen-AI assisted changes", report claims. Tom's Hardware. Available: https://www.tomshardware.com/tech-industry/artificial-intelligence/amazon-calls-engineers-to-address-issues-caused-by-use-of-ai-tools-report-claims-company-says-recent-incidents-had-high-blast-radius-and-were-allegedly-related-to-gen-ai-assisted-changes. Accessed 2026 Apr 15.

[5] BetterUp Labs (2025) Workslop: The hidden cost of AI-generated busywork. BetterUp. Available: https://www.betterup.com/workslop. Accessed 2026 Apr 15.

References & Reading

[6] Alpert S (2024) Everyone is wrong about that Slack flowchart. Sophie Alpert. Available: https://sophiebits.com/2024/10/30/everyone-is-wrong-about-that-slack-flowchart. Accessed 2026 Apr 15.

[7] Steinberger P (2024) Just talk to it. steipete. Available: https://steipete.me/posts/just-talk-to-it. Accessed 2026 Apr 15.

[8] Niederhoffer K, Robichaux A, Hancock JT (2026) Why People Create AI "Workslop"-and How to Stop It. Harvard Business Review. Available: https://hbr.org/2026/01/why-people-create-ai-workslop-and-how-to-stop-it. Accessed 2026 Apr 15.

[9] Cowan RS (1983) More Work for Mother: The Ironies of Household Technology from the Open Hearth to the Microwave. New York: Basic Books.

[10] Wajcman J (2018) Digital technology, work extension and the acceleration society. German Journal of Human Resource Management. Available: https://journals.sagepub.com/doi/10.1177/2397002218775930. Accessed 2026 Apr 15.

[11] Nadh K (2026) Code is cheap. Show me the talk. nadh.in. Available: https://nadh.in/blog/code-is-cheap/. Accessed 2026 Apr 15.

[12] Bedard J, Kropp M, Hsu M, Karaman OT, Hawes J, Rosen Kellerman G (2026) When Using AI Leads to "Brain Fry". Harvard Business Review. Available: https://hbr.org/2026/03/when-using-ai-leads-to-brain-fry. Accessed 2026 Apr 15.

[13] Cherny B (2026) Post on X. Available: https://x.com/bcherny/status/2022762422302576970. Accessed 2026 Apr 16.